LIGHTS BACK ON

Development roadmap
and complete design blueprint

SOLO + 1-4 PLAYER CO-OP / WINDOWS STEAM

36-week planning baseline. 66 ordered work items.
12 contract sheets. 12 screen groups.
46 planned acceptance cases. 5 visual references.

VERSION 1.0 / 27 OCTOBER 2026

Production specification, not a completed game.
Estimates and targets remain subject to measured validation.

How to use this blueprint

Build the power puzzle and prove real Steam connectivity before expanding the content. Start with F01 in the task backlog. Finish each acceptance check and retain evidence before advancing its dependency gate.

The master blueprint defines the product and technical rules. Contract sheets make all twelve missions concrete. Screen specifications describe normal and failure states. The backlog is ordered work; the acceptance matrix is the release gate.

Only the included Python reference capacity/topology tests were run for this package. The game, Godot integration, Steam connection, device performance, and demand tests are still future work.

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1 / Master blueprint

Outcome: a Windows Steam game in which one to four players restore a storm-damaged neighborhood. The distinctive mechanic is an interconnected power network: diagnosing faults, choosing temporary supply routes, managing limited capacity, and seeing buildings return to life.

Status: production design and planning package. This is not a finished Godot game. Screens are specification mockups. The included Python reference model verifies the proposed circuit rules and first mission; it does not verify engine integration, networking, performance, player demand, or commercial viability.

Evidence labels: VERIFIED PLATFORM FACT means an externally documented capability or requirement. DESIGN DECISION means the chosen implementation. TARGET means a value to test. ESTIMATE means planning capacity, not a delivery guarantee. No sales or wishlist forecast is assumed.

1. Product decisions to lock

Area	Implementation decision
Players	Solo and 1-4 online players; friends-only Steam lobbies first
Perspective	First person; visible tool and gloved hands; readable player silhouettes
Tone	Stormy, warm, practical, occasionally funny; weather and mistakes create pressure
Platform	Windows x64 at release; controller and keyboard/mouse supported
Engine	Godot 4.7.2 standard, typed GDScript; Compatibility renderer baseline
Session	Target 20-30 minutes with a group, 25-40 solo; measure in testing
First release	Three compact districts, six service locations each, four contracts per district
Demo	Alder Street, contract C01; one complete mission and its built-in tutorial
Presentation	Stylized 3D, reusable modular architecture, warm restored windows against cool rain
Business model	Premium purchase; proposed US list price $12.99, to validate after the demo
Progression	Contract unlocks, toolkit capacity, van/depot customization; all essential tools available initially
Save model	Host owns campaign world; every participant gets personal completion and reward receipts

VERIFIED PLATFORM FACT: the official Windows download page lists Godot 4.7.2 as stable at preparation time [S1]. That does not establish compatibility with a particular Steam extension binary. Step F02 must qualify and lock the engine, export templates, GodotSteam integration, multiplayer peer, and Steamworks runtime as a tested tuple.

The release scope excludes open-world driving, public matchmaking, host migration, simulated rope physics, terrain deformation, real fluid simulation, procedural city generation, NPC schedules, combat, and a custom online backend. These are explicit production boundaries. A depot loading transition replaces driving between districts. Weather is authored VFX and timed state changes. Flooding uses a water-height parameter and access checks.

2. What makes the game worth playing

Three pillars control every feature decision:

1. **Understand a fault.** A meter gives enough evidence to infer a cause. A marker may identify a building, but it must not reveal the exact repair immediately.
2. **Make a useful tradeoff.** A temporary generator helps one branch but cannot supply everything. Opening a route, moving supply, or shedding optional load changes the next available action.
3. **See and hear recovery.** Windows, signs, fans, music, and residents' short reactions confirm a meaningful result. Network recovery must remain legible without darkness hiding the scene.

The loop is: choose contract -> load van -> arrive -> inspect -> isolate fault -> repair or reroute -> verify supply -> secure site -> receive results -> upgrade -> choose next contract.

The system must stay interesting when players communicate efficiently. Repeating repair animations is not the intended source of difficulty. If testers can solve every mission by following markers and holding Interact, revise the fault layout before adding content.

First five minutes of C01

Time target	Player action	Implementation and feedback
0:00-0:30	Step out of van	Main street dark; lit van and reflective junction labels establish orientation
0:30-1:15	Test west junction	Meter shows healthy supply but tripped branch; optional training hint identifies isolation control
1:15-2:15	Isolate and replace west fuse	One turn-to-off action, two tool presses, one replacement snap; no wire-color trivia
2:15-3:00	Reset west branch	Apartments and market illuminate in a short stagger; map gains a verified supply path
3:00-4:15	Power basement pump	Carry generator, connect one socket, choose pump branch; water-height bar starts falling
4:15-5:00	Choose next task	Pub remains dark; teammate can inspect east branch while pump exposes trunk repair

These are pacing targets rather than recorded player timings.

3. Release content and contract progression

The twelve authored contracts are the campaign. District geography is reused intentionally; each contract changes faults, supply availability, route access, and optional priorities. After a contract is completed, curated seeded variants may be replayed. A seed must choose from validated templates, never arbitrary unsolvable damage.

ID	District / contract	New decision or mechanic	Prerequisite
C01	Alder Street / First Shift	Diagnose, isolate, fuse repair, generator, pump access	None
C02	Alder Street / Dinner Rush	Market and pub compete for temporary supply	C01

ID	District / contract	New decision or mechanic	Prerequisite
C03	Alder Street / Roof Route	Fixed access ladder and alternate overhead connection	C02
C04	Alder Street / The Whole Block	Two faults plus an optional service overload	C03
C05	Canal Yard / Rising Water	Two gated repair locations; pump ordering	C04
C06	Canal Yard / Cold Chain	Cold-store grace timer; temporary power scheduling	C05
C07	Canal Yard / Wrong Label	Conflicting label and actual meter evidence	C06
C08	Canal Yard / Split Supply	Two mutually exclusive source routes and a tie switch	C07
C09	Hill Road / Wind Warning	Authored gust window closes a rooftop route temporarily	C08
C10	Hill Road / Community Night	Optional event loads create a clear capacity choice	C09
C11	Hill Road / Borrowed Power	Generator relocation while preserving telecom supply through a bypass	C10
C12	Hill Road / Lights Back On	Combined systems; final district-wide restoration	C11

Six reusable location archetypes are housing, food/retail, social venue, utility room, workshop/storage, and public service. Eighteen locations use those kits with distinct signs and silhouettes. Only the task areas need interiors: utility rooms, accessible shop areas, a small basement, and selected roof landings. Most homes remain exterior facades.

Contract generation records: contract ID, template revision, seed, required loads, optional loads, initial faults, tool allowances, event timeline, route predicates, reward definition, and one known-valid solution trace. A content validator rejects missing IDs, unreachable objectives, absent required tools, circular prerequisites, and final demand above available capacity.

4. First district layout

Alder Street occupies a proposed 110 m x 70 m playable envelope. One meter equals one Godot unit. The street forms a navigable loop with a rear service alley. Use two broad sightlines so players can see recovered buildings from the van and east junction.

Location	Approximate center (x,z), meters	Purpose
Van / spawn	(8, 9)	Resupply, briefing, extraction, emergency recovery
West junction JW	(30, 24)	First diagnosis and fuse tutorial
Apartments west	(20, 46)	First conspicuous restoration reward
Market	(43, 46)	West branch load and optional cold cabinet
Pub	(70, 46)	East branch, jukebox, optional venue load
Apartments east	(92, 46)	Main east restoration objective
Workshop	(88, 18)	East branch load and service alley landmark

Location	Approximate center (x,z), meters	Purpose
Pump house / basement	(60, 16)	Generator decision and east trunk access
East junction JE	(73, 24)	East repair and overload demonstration

Place interactive panels at 1.2-1.6 m height. Make primary circulation at least 2.4 m wide and door clear width at least 1.1 m. Use a 4 m structural module and 0.25 m placement increments. Main route slopes should allow the same CharacterBody3D movement used on stairs; test controller traversal before art polish. A fixed ladder is a short interaction transition, not a free physics object.

Collision is simplified convex geometry. Handrails, cables, small debris, and grass do not snag players. A misplaced carried item returns to the nearest valid drop anchor; an out-of-bounds recovery action is always available from the van menu. A player recovery returns that player to the van without resetting the network.

5. Exact power-network rules

This is a deliberately abstract capacity puzzle. Power units (PU) are invented gameplay units. Do not represent the model as an electrical engineering simulator.

Data and topology

Nodes represent sources, junctions, and consumers. An edge is a directed potential supply path with an enabled flag, health flag, PU capacity, and stable ID. A junction can be isolated, have a failed component, or have a tripped breaker. Each consumer has a PU demand and an enable switch. Each active consumer may have at most one energized incoming route. A consumer may have alternate potential routes only when at most one is electrically active.

The energized graph is a collection of directed trees, one source per connected component. Reject cycles, source-to-source connections, and any two simultaneously energized incoming routes. Invalid topology is refused before state changes, with the map highlighting the conflicting tie. This avoids ambiguous source sharing and makes diagnosis explainable.

Evaluation order, fixed host tick

1. Validate IDs and allowed topology; sort by stable entity ID for all traversal.
2. Remove disabled/broken edges and isolated, failed, or tripped junctions from the active paths.
3. Traverse outwards from each online source. Mark reachable healthy consumers as provisionally supplied.
4. Sum enabled reachable consumer demand back towards its source. Record the downstream PU on every active edge.
5. Compare each source and edge against its capacity. A value strictly greater than capacity overloads; equality is allowed.
6. Accumulate each overloaded protected branch's timer at 10 Hz. At 1.5 continuous seconds of overload, trip its designated upstream breaker. Healthy demand resets its timer to zero.
7. Recompute after trips. A reset is accepted only after demand and topology are valid; a refused reset does not start repeated trip cycles.
8. Emit authoritative state changes. Clients animate effects from those changes; client visuals never decide whether a building is supplied.

The Python reference evaluator in this package implements steps 1-5 and topology validation. It reports which capacities are exceeded. The timed breaker, repair state machine, network authority, and visual effects are specified here and must be implemented and tested in Godot; the reference evaluator does not claim to implement them.

C01 starting network

Source / branch	Capacity / demand	Initial condition
Grid source	24 PU	Online
Grid -> JW	12 PU	West fuse failed; breaker tripped
Grid -> JE	12 PU	East trunk cable broken
JW -> apartments west	4 PU demand	Healthy consumer
JW -> market	5 PU demand	Healthy consumer
JE -> pub	4 PU demand	Healthy consumer
JE -> apartments east	4 PU demand	Healthy consumer
JE -> workshop	3 PU demand	Healthy consumer
Portable generator -> TEMP	6 PU	Initially off
TEMP -> pump	4 PU demand	Initially disconnected; required access task
TEMP -> pub alternate	4 PU demand	Optional temporary route; default disconnected
JE -> pub sign / test load	3 PU demand	Training load; default disabled

The west branch totals 9 PU. The east required branch totals 11 PU. The grid final required demand is 20 PU. The generator can power the 4 PU pump or the 4 PU pub, but not both together. The pub must never have the temporary and grid routes energized simultaneously. Closing the 3 PU sign test load on the east branch produces 14 PU against its 12 PU edge limit, despite the grid source itself still having spare capacity. This teaches branch limits rather than merely source limits. In C01 the sign is a training load, excluded from optional-service bonuses and star requirements; leaving it off is correct. The generator has no fuel depletion in this contract.

Known-valid C01 solution

1. Inspect JW. Open its isolator. Replace its fuse while isolated. Close isolator and reset the breaker. West apartments and market receive 9 PU.
2. Place the generator at the pump pad. Connect only TEMP -> pump. Start the generator. After 12 continuous supplied seconds, basement water reaches the access threshold.
3. The pump task latches complete. A flap valve holds the drained basement for the rest of this contract; switching off the pump does not refill it. This deliberately prevents a resource deadlock.
4. Isolate JE, repair the exposed east trunk, then restore JE. Required east loads receive 11 PU. Keep TEMP -> pub disconnected.
5. Disconnect and stow the temporary generator. Verify all five required building consumers have grid supply for 10 continuous seconds.
6. Stow deployed equipment at any marked storage anchor. All players currently connected enter the extraction zone; the host starts a five-second departure countdown. Any player can cancel during that countdown.

The pump is an access requirement rather than a final continuously supplied consumer. The sign test load does not affect completion. A recovery action can return the generator to the van; it cannot remove a permanent repair or consume irreplaceable parts.

6. Player controls and interactions

Action	Keyboard / mouse default	Controller default
Move / look	WASD / mouse	Left / right stick
Interact / confirm	E	South face button
Tool primary	Left mouse	Right trigger
Tool secondary / rotate	Right mouse	Left trigger
Tool selection	1-4 or wheel	Hold right bumper for radial
Drop carried item	G	Hold west face button
Map / meter details	Tab	View / back
Context ping	Middle mouse	Left bumper
Sprint / crouch	Shift / Ctrl	Left / right stick click
Pause / leave menu	Escape	Menu / start

All gameplay bindings are remappable. Controller glyphs follow the active input device. Standardized south/east/west/north descriptions in design become platform glyphs in game. Maintain a complete focus path through every menu.

Character targets: 4.0 m/s walk, 5.5 m/s sprint, 2.8 m/s while carrying the generator, 1.75 m camera height, 80-degree default horizontal FOV with 65-105 slider. These are starting values. Sprint has no stamina system. The generator can be moved solo or on a wheeled cart; no mandatory two-person carry exists.

An interactable is selected by a 2.3 m center ray plus a small aim-assist cone. Prompts name the object, verb, and missing condition: `East junction | Repair cable | Isolate branch first`. A status strip on the object and an on-screen icon repeat its state. Invalid interactions explain one specific reason and do not consume supplies.

Repair state machine

Undiagnosed -> diagnosed -> isolated -> opened -> component removed -> replacement installed -> closed -> tested -> restored. Canceling midway leaves the last durable state intact. The repair has discrete presses or short turns; no uninterrupted hold lasts more than 1.2 seconds by default. The accessibility setting can replace repeated presses with a single confirmation.

One player may own a repair lock at a time. Other players see who is working and can inspect or ping. The host releases that lock when its owner moves out of range, cancels, disconnects, or sends no valid repair heartbeat for two seconds. Tool availability and component consumption are checked by the host on each durable transition.

The three base faults are component failure, broken connection, and overload/tripped breaker. The meter shows source presence, continuity, branch demand, capacity, and breaker reason. Unknown data uses a question mark, not a false zero. Repairs on an unisolated panel are refused with feedback. Mistakes cause recoverable trips or a short stumble; there is no mandatory death/revive loop.

7. Cooperation, solo mode, and communication

Roles are tasks players choose, not permanent classes: diagnose, repair, move supply, check the map. Every player can use every tool. Solo mode uses the identical network rules, reduced optional pressure, one-person equipment handling, and paused simulation while the local pause menu is open. A multiplayer pause menu never stops the host world.

Use four context pings: fault here, need supply, isolate here, ready to test. Each includes location ID, icon, color, and short localized text. Limit each player to one active directional ping and four new pings per two seconds. Pings expire after eight seconds or when their condition is resolved.

Provide canned radial messages and a compact text chat. Text has a maximum length, rate limit, markup escaping, mute controls, and no URL launching. Integrated voice chat is deferred; the game must be completable with its pings alone. Do not advertise built-in voice. This saves a separate audio networking and moderation feature while leaving the game fully cooperative.

Ready checks require every connected player to respond; the host can remove an unresponsive player from the lobby. On completion, any connected player can trigger `Ready to leave`, but the host controls departure. A host leaving ends the session with a clear saved-checkpoint message. Host migration is not part of version 1.0.

8. Technical implementation and file ownership

VERIFIED PLATFORM FACT: Godot exposes high-level multiplayer, RPC authority, reliable/unreliable modes, and sender identification [S2]. Steam lobbies and Steam networking are separate concerns [S4]. The design below selects how to use them; it does not assume a lobby itself transports gameplay.

Module / proposed path	Owns	Must not own
`autoload/game_session.gd`	Session state and scene transitions	Grid calculations or UI drawing
`network/transport_adapter.gd`	Offline, ENet development, Steam peer selection	Mission rewards
`network/session_host.gd`	Admission, input validation, snapshots, entity IDs	Rendering
`simulation/power_grid.gd`	Pure graph evaluation, capacities, trip timers	Steam calls or scene searches
`simulation/mission_director.gd`	Objectives, events, route predicates, extraction	Player-owned progression
`interaction/interaction_service.gd`	Reachability, locks, state transitions	Direct client-authoritative repairs
`actors/player_controller.gd`	Local input, predicted motion, interaction requests	Reward issuance
`world/service_location.gd`	Building presentation from supplied state	Whether the supply is valid
`persistence/save_service.gd`	Versioned saves, backup, atomic replacement	Broad Cloud configuration
`progression/reward_service.gd`	Completion receipt IDs and deduplicated rewards	Rendering or transport

Module / proposed path	Owns	Must not own
ui/	Views, presenters, focus and input hints	Changes to simulation state without commands
content/	Typed definitions and imported JSON resources	Executable downloaded scripts

Root directories: scenes/, scripts/, content/, assets/, tests/, tools/, docs/. Keep original art source files outside the exported Godot asset tree. Use Git LFS for large originals. Do not commit exported builds or local save data. No repository is created or modified by this planning package.

Scenes: Boot, MainMenu, Depot, Lobby, MissionRoot, PauseOverlay, Results. MissionRoot contains NetworkRoot, DistrictRoot, Players, Equipment, MissionDirector, and HUD. Stable object IDs come from authored content, never scene-tree order. Every peer builds the same gameplay node paths before accepting RPCs. Additive cosmetic nodes must not change gameplay RPC paths.

Host authority and command protocol

The host is authoritative over grid state, repairs, equipment placement, objectives, mission clock, and rewards. Clients submit intentions: inspect, request_lock, repair_step, toggle_switch, place_item, ping, ready, request_extract. Each request includes a session ID, sequence, entity ID, and bounded payload. The host obtains the sender identity from the transport, validates membership, distance, state, inventory, lock ownership, rate, and content version, then emits a result with a revision.

The first prototype targets 30 Hz host movement simulation, 15 Hz movement snapshots, and 10 Hz grid evaluation. Reliable ordered messages carry durable interactions, spawns, objectives, and results. Unreliable sequenced updates carry movement. Use transport channels only after the chosen Steam peer demonstrates channel support. A single-channel reliable control path is the compatibility baseline; do not copy ENet channel assumptions into a plugin without testing.

Local movement is predicted and reconciled against host state; remote actors interpolate approximately 100 ms behind. Equipment has server-owned placement anchors and cosmetic smoothing. Cables are authored curves between sockets, not replicated rigid-body ropes. Constrain loose props to a small approved set; no gameplay depends on their exact physics state.

On joining or reconnecting, freeze the new player in a joining screen, send a snapshot manifest and bounded chunks, verify contract/content hashes, apply the snapshot, then process queued deltas newer than that snapshot revision. Deduplicate commands by session and sender sequence. A reconnecting player may rejoin the current mission while the host remains; their carried object has already returned to a safe pad. Disconnected participants are not required for extraction.

Network test targets: 80 ms and 150 ms round-trip latency, 1% and 3% packet loss, jitter up to 30 ms, ten-minute connection soak, 20 repeated lobby transitions, and a four-player 60-minute soak. Use controlled impairment at a test router or network harness and record the actual configuration. A local multi-instance test does not satisfy the real-network gate.

Save and progression contract

Use user://profiles/<account_key>/campaign.json, profile.json, receipts.json, and checkpoints/<campaign_id>.json. Write to a temporary file in the same directory, flush and close, keep one previous valid backup, then atomically replace the target. Include schema_version, build_id, content_revision, UTC timestamp, and stable record IDs. A checksum detects corruption; it is not an anti-cheat system.

Save host checkpoints every 60 seconds and after durable repair completion. Loading recomputes derived power state from saved causes, not from visual light states. Clients save only personal profile and receipts. A completion receipt key is campaign_id + contract_instance_id + account_key; rewards are applied once. Host-local operation follows the same command and reward path as a remote client.

Guest completion unlocks the corresponding contract in that guest's personal board, and all earlier contracts, to prevent replaying already-completed prerequisites. It does not overwrite a separate host campaign's active repair state. Credits are personal, never a shared spendable balance. A guest's unlocked tools are cosmetic/convenience loadout choices; every contract provides its required equipment even to a new guest.

Steam Auto-Cloud should sync campaign, profile, and receipt files only; machine-specific graphics, bindings, and logs stay local. Valve documents both Cloud API and Auto-Cloud approaches [S5]. Demo saves use a separate demo namespace. If demo progress import is later enabled, import only a versioned personal completion record, never overwrite a full-game campaign.

9. UI/UX implementation

The companion screen specification defines all twelve screen groups, states, focus order, and primary actions. Five 1920x1080 images show the field HUD, network tablet, depot, lobby, and district plan. They are exact layout references, not final game renders.

Design tokens: background `#111B25`; panel `#1B2A38`; elevated panel `#243748`; primary text `#F3F5EF`; muted text `#B9C6CF`; restored teal `#65D6B4`; action amber `#FFC86B`; fault coral `#F28B82`; unknown gray-blue `#7E93A3`. Status always includes an icon and text. Confirm contrast on the actual rendered background, including translucent overlays.

Use Noto Sans or another explicitly licensed font with required language coverage. Body text starts at 22 px on the 1920x1080 design canvas, primary button text at 26 px, headings 32-44 px. Minimum click target 48x48 px. Use an 80 px safe area and a UI scale slider from 80% to 150%. Layout containers reflow rather than clipping when text grows. Support 16:9 and 16:10 first; test 21:9 with bounded centered UI.

On-screen hierarchy: mission objective top-left; crew status top-right; world at center; context prompt below reticle; equipment bottom-left; latest noncritical event bottom-right. Keep at most three objective rows visible. Breaker trips receive a short warning and a persistent map icon, not a full-screen failure panel. Major restoration receives a 1.2-second audio/lighting reward, without blocking input.

Accessibility settings include separate mouse sensitivity/inversion, remapping, dead-zone sliders, FOV, head-bob disable, camera-shake disable, reduced flashes, subtitles and speaker labels, UI scale, hold/toggle options, color-independent status icons, and separate audio buses. Reduced-flash mode replaces arcs with a steady amber indicator. No task uses hearing, color, or voice alone.

English and Simplified Chinese interface and subtitles are the release language target. All text uses localization keys from the first UI pass. Keep strings out of images and textures. Test a 35% expansion pseudo-locale and a Chinese font before locking panels. Translation and terminology review is an explicit backlog task; it is not assumed completed by this English plan.

10. Visual art, animation, audio, and effects

Art direction

Rounded but grounded architecture, readable silhouettes, reflective workwear, chunky tools, and limited surface noise. Cool blue street lighting contrasts with warm cream recovered windows. A wet surface is a material response plus decal highlights, not an expensive full-screen reflection requirement. The dark state still preserves navigation through sky fill, reflective labels, and the van.

Build one 4 m modular architectural kit: wall, window wall, door wall, corner, roof edge, roof plane, floor, stair flight, railing, service panel recess. Six facade palettes create the location archetypes. Required props include three junction designs, a generator, cart, fuse case, cable reel, meter, toolbox, isolation lever, work light, pump, van exterior/interior station, road barriers, drain, signs, bins, and limited debris.

Three visible states per service location are dark, temporary supply, and restored grid supply. Temporary supply uses selected lights and a small socket indicator, making it visibly different from full restoration. Store light groups as node references or exported arrays, not repeated runtime tree searches. For a large window

array, emissive materials provide most of the effect; only selected practical lights cast shadows.

Character scope: one shared worker body/rig with four helmet/vest palettes and a neutral silhouette. First-person hands use that rig where practical. Required animations: idle, walk, sprint, crouch, carry, interact, tester use, component replacement, switch operation, stumble, recovery, and ping gesture. No facial animation or lip sync. A hand IK target aligns tools to panels; verify contact before adding secondary animation.

Starting performance budgets

These are internal caps to profile, not published minimum specifications: 250 visible draw calls in a representative street view, 800k visible triangles, 512 MB loaded texture budget for the mission, two concurrent shadow-casting local lights plus the main directional light, and 24 simultaneously audible 3D voices. Atlas small props; use 1k textures for ordinary props and 2k for large shared kits. Texture choices must be confirmed by on-device profiling.

Target the user's GTX 1650 4 GB class machine at 1920x1080 low/medium, 60 fps, using an exported four-player host build. Record actual frame times, VRAM, system RAM, CPU, drivers, and resolution. The target is unverified until that machine or an equivalent controlled system is measured. Provide presets, a render-scale control, and independent rain/shadow quality options.

Audio and effects implementation

Audio buses: master, ambience, tools, UI, music, dialogue. Build a rain loop, sheltered-rain variant, wind layer, generator idle/load layers, tester click/beep, switch clunk, fuse action, breaker trip, pump start/drain/stop, cable snap, footsteps for three surfaces, ping sounds, and a restoration chord. Layer restored venues into ambience at bounded volumes. The pump has visual state and a subtitle cue as well as audio.

Use a small set of voiced radio barks only if budget allows. Text-only radio subtitles are the release fallback. Every essential instruction already exists in UI text and ping icons. Music is sparse during work and resolves briefly at restoration. Normalize loudness across tool actions; do not make shock sounds disproportionately loud.

Weather uses screen-space rain particles near each player, roof drip emitters, distant lightning without rapid flashes, and authored gust events. Water height is a scalar interpolated on a plane mesh. Do not simulate volume flow. Each electrical effect is driven by the host's fault state and may be locally reduced for accessibility.

11. Economy and replay design

The proposed starting economy is intentionally simple. Contract base reward is 500 credits, completion bonus 100, optional-service bonus 50 each capped at 100, and equipment-restowed bonus 50. There is no negative balance or repair bill. A poor run still grants the base reward on successful completion. Restarts cost nothing. Reward numbers are balance inputs, not financial values.

First upgrades: extra cable storage 600, brighter tester screen 450, second work light 350, cart storage mount 800, and van palette 300. Essential repairs never require a purchased upgrade. Offer sidegrades and organization improvements before raw speed; an experienced player should not make a newcomer irrelevant.

Scoring reports required services restored, optional services, trips caused, crew recoveries, time, and equipment collected. Stars are: one for completion, a second for all optional services, a third for no preventable branch trips after the tutorial warning. There is no speed-based star, avoiding an implicit penalty for accessibility settings or careful diagnosis. Timed contracts affect optional bonuses, not total campaign access.

Replay variation comes from fault location, service demand choices, access prerequisites, and one or two authored weather events. Never change fault position invisibly during a diagnosis. A warning event may damage a new branch only with advance readable cues. Every generated variant must retain a known solution and a recovery route.

12. Roadmap, capacity, and milestone gates

ESTIMATE: 36 production weeks, with one experienced full-time gameplay/network programmer at 30 focused hours per week, one art/UI generalist at 20 hours per week, and external playtest support. That is approximately 1,080 engineering hours and 720 art/UI hours before contractor QA and contingency. Reserve a further 6-8 weeks if the Steam networking spike, content throughput, or device performance fails its first gate. The estimate has not been calibrated against this team's actual velocity.

If one person does all disciplines at 30 focused hours per week, the same nominal 1,800 hours is 60 focused weeks before additional context-switching or contingency. Do not present a multiplayer commercial game as a guaranteed few-week build.

Milestone	Planned weeks	Ordered work	Gate / required evidence
F: Foundation	1-2	Repository, engine/Steam spike, exports, input, build manifest	Two PCs on separate networks connect, move, toggle a shared switch, reconnect
G: Grid toy	3-5	Pure graph evaluator, faults, meter, isolation, one consumer, save skeleton	Deterministic rules, no invalid source merges, all fault cases pass
V: Playable slice	6-9	Alder graybox, C01, generator/pump, two then four players, recovery	Complete C01 solo and with four; no critical blocker in ten runs
A: Validation	10-12	Art sample, sound reward, UI, external testing, measurement	Fun and comprehension gate below; choose expand/revise/stop
D: Public demo	13-16	Polished C01, Steam store assets, localization pass, demo QA	20 fresh tester sessions, stable invites, clean-machine install
C: Content	17-24	District kits, C02-C12, tool variants, economy, seed templates	All contracts solvable; per-contract solution trace and regression evidence
H: Alpha	25-28	Feature freeze, accessibility, controller, persistence, optimization	Entire campaign complete solo/online; no known save-loss bug
B: Beta	29-32	External network/device tests, Cloud, localization review, exploit fixes	Release matrix passed; no P0/P1 defects; stable economy receipts
R: Release	33-36	Store/build review buffer, launch candidate, trailer, release checklist	Steam approvals, exact candidate hash, final smoke, rollback build retained

Milestones are sequential gates; weeks are planning slots. A failed gate moves the schedule. Do not fill the next district with art while the current interaction or network model needs redesign. See `02-Implementation-Backlog.json` for task-level inputs, dependencies, outputs, effort, and acceptance criteria.

Validation gate after the slice

Recruit at least 12 people outside development, arranged into six two-player sessions, and separately run at least three solo sessions. Record first-repair time, wrong actions, unassisted completion, idle time, and a post-session 1-5 desire-to-play-again score. Obtain permission for any recording. No tester needs to publish a review or endorse the game.

Proposed continuation thresholds are: at least five of six pairs finish without developer instructions; median pair first repair below four minutes; at least eight of twelve players rate desire to play again at four or five; and the median pair voluntarily accepts an offered second variant. These are internal decision rules, not proven industry benchmarks and not statistical evidence of market demand. If the sample is recruitment-biased or most people are friends, state that in the report.

If comprehension fails, change meter feedback and signposting. If replay desire fails despite comprehension, change the tradeoffs and fault combinations. If four-player sessions leave players idle, add parallel tasks or shorten the mission; do not just multiply repair health. A second failed slice test triggers a scope/premise review before production spend.

First ten working days

Day	Work in exact order	End-of-day proof
1	Create project/repository; install stable engine + matching templates; configure actions; export blank Windows build	Clean checkout imports and launches; version/hash manifest recorded
2	Create adapter boundary; integrate candidate Steam wrapper/peer; add lobby creation/join and protocol revision	Two separate licensed test accounts see the same private test lobby
3	Add shared player spawning, one authoritative switch, command sequence, rejection response	Switch revision agrees on both real PCs; duplicate request changes state once
4	Test separate-network invite, disconnect/rejoin, wrong version, failed Steam initialization	Recorded pass/fail report; exact compatible tuple pinned or gate remains blocked
5	Implement pure power evaluator using this package's reference cases	Unit tests match the reference outputs; no dependency on scenes
6	Build a one-room panel and consumer; wire inspection and isolation	Meter changes from unknown to measured; live repair correctly refused
7	Implement fuse states, lock ownership, inventory consumption, restored-light event	Two players cannot consume the same fuse; cancel/resume preserves state
8	Implement generator and two consumers, demand meter, overload delay	4 PU on 6 succeeds; 8 PU trips after 1.5 s; invalid reset is refused
9	Connect the small loop to host commands; create versioned save/checkpoint	Reload restores causes and recomputes supply; duplicate reward is ignored
10	Export, run clean-machine and second-network tests, record five-minute raw capture	Demonstrable multiplayer grid toy; written failures become blocking tasks

Do not skip the Steam transport spike and substitute a localhost-only demonstration. If Steamworks access is not yet ready, proceed with the pure grid toy in parallel, but mark the real-network gate pending and do not claim the co-op architecture qualified.

13. Build, QA, and release workflow

Every task uses a short-lived branch. Review the diff, run relevant checks, merge to an integration branch, and export milestone builds from a tagged revision. Include `build_manifest.json` with engine version, export-template hash, extension hashes, content revision, Git commit, protocol version, and save-schema version. Reproduce one export from a clean checkout before each external milestone.

Use a plain internal build archive initially. Steam branches become `internal`, `playtest`, and `release-candidate` once the app is configured. Demo and full game have separate content manifests. Debug tools are available only in internal exports. The release pipeline fails if production resources reference placeholder or unlicensed assets.

The acceptance matrix covers simulation, simultaneous commands, disconnects, recovery, saves, controller navigation, localization, access settings, and exported performance. Test the authoritative state, not merely the apparent lights. Compare final canonical grid hashes across peers, excluding position and visual interpolation data. Capture P95 frame times and profiler traces rather than reporting only average fps.

Defect severity: P0 = save loss, arbitrary file/code access, or launch failure; P1 = mission cannot finish, repeatable disconnect, duplicate spend/reward, or essential input inaccessible; P2 = incorrect feedback with a workaround; P3 = cosmetic. No known P0/P1 issue is accepted into the release candidate. P2 exceptions need a specific impact statement and fix decision.

Suggested on-device target: P95 total frame time below 20 ms in the representative stress route, with a 60 fps cap enabled; no sustained VRAM growth across five mission reloads. Treat 60 fps and that frame-time threshold as internal targets until measured. Avoid publishing hardware minimums before beta measurements.

Steam-specific work

1. Complete Steamworks account/app setup and establish authorized test accounts. Record the app IDs privately. A public sample app ID may support an early technical experiment but is never a production or isolated playtest namespace.
2. Configure Windows executable, depots, branches, launch options, ownership packages, and demo app. Test a downloaded branch build rather than only a local editor run.
3. Produce capsule art, library art, screenshots, trailer, descriptions, tags, content survey, support contact, and language declarations. Use the current official asset templates and dimensions [S8]. No Steam logo or wishlist claim is baked into the game logo.
4. Publish Coming Soon after the slice can supply representative footage. Treat the public name as provisional until name/trademark checks are complete; no name clearance is claimed here.
5. Publish a stable demo only after the demo gate. Steam Next Fest permits one participation per title under its current rules and requires an unreleased base game, public page, and playable demo [S7]. Select the edition after the stable demo exists; use that edition's published deadlines.
6. Submit store presence and near-final build with at least seven business days of review buffer for each relevant submission. Valve says typical review is 3-5 business days and recommends extra time [S6]. Recheck current rules at release preparation.
7. Respect applicable Steam waiting periods. The current Steam Direct page describes a 30-day fee-to-release wait and a Coming Soon page visible for at least two weeks [S9]. Those are not substitutes for account, tax, review, or event readiness.
8. Verify achievements, Cloud sync, invites, overlay, offline solo behavior, and ownership/version errors using the actual release candidate. Exclude machine graphics settings from Cloud.
9. Stage release, retain the previous approved build, and record the exact final manifest. Publishing is a separate operational action; this planning package does not publish anything.

Marketing deliverables are one 45-60-second gameplay trailer, six representative screenshots, a short store description, three feature panels, one clear demo announcement, and a press folder. Trailer order: dark street (0-3 s), diagnosis (3-10), cooperation (10-20), overload consequence (20-28), alternate solution (28-40), full restoration (40-50), title/action (50-60). Avoid a long studio-logo opening.

14. Measurement, risk controls, and cut order

Collect local anonymous event logs during development: session start/end, contract/version, objective transitions, branch trips, first repair, hint activation, idle interval, recovery, disconnect, and completion. Use random session IDs, not raw Steam IDs, in analytics exports. Uploading telemetry requires a separate implementation and privacy decision; the baseline kit writes local logs. Do not conflate tester feedback, demo analytics, wishlists, and paid demand.

Risk	Early test	Response if it fails
Steam transport compatibility	Week 1-2 two-PC test	Pin a documented working tuple; stop content expansion until proven

Risk	Early test	Response if it fails
Repetitive repairs	Week 10-12 replay test	Reduce repair clicks; add supply/access decisions
Four-player idle time	C01 activity timeline	Parallelize west repair, pump setup, east diagnosis; shorten backtracking
Low-end lighting cost	Art sample on GTX 1650 class	Favor emission, fewer shadow lights, bounded rain and reflections
Save/receipt inconsistency	Crash/rejoin/replay tests	Centralize durable transitions and receipt deduplication
Content throughput too slow	Time C02 production	Reuse geometry; reduce contracts before sacrificing QA
Weak player interest	Observed demo behavior + interviews	Revise premise/scope before full content spend; no assumed sales rescue

Cut order if capacity is exceeded: voiced barks -> cosmetic variants -> seeded replay variants -> reduce contracts 12 to 8 -> reduce districts 3 to 2. Preserve the power puzzle, recoverable failures, solo support, functioning co-op, save integrity, and readable feedback. Removing co-op changes the commercial proposition and requires a fresh product decision, not an unnoticed scope cut.

After every milestone, record actual engineering/art hours, asset throughput, defect count, unresolved risks, and the next estimate. Replace the 36-week baseline with measured velocity after the slice. Use no public release date until the beta gate is passed.

15. Package contents and exact use

Start with `README.md`, then this blueprint. Open `design/` for five layout references, `03-Screen-Specifications.md` for screen behavior, `04-Acceptance-Test-Matrix.md` for checks, and `02-Implementation-Backlog.json` for dependency-ordered work. `05-Content-and-Asset-Manifest.json` lists contracts, assets, tools, and audio. `06-Contract-Design-Sheets.md` specifies all twelve missions, including their supply limits, starting faults, optional objectives, and solution sequences. `data/first_shift.json` is the sample mission definition.

The reference model has no third-party Python dependencies. From the extracted package root run `python -m unittest discover -s tests -v`, then `python reference/power_model.py data/first_shift.json`. It evaluates the initial and authored solved configurations and reports the deliberate overload cases. The output does not mean the game is implemented.

The production order is F -> G -> V -> A -> D -> C -> H -> B -> R. Pick the first task whose dependencies are complete. Finish its acceptance check and attach the evidence before moving it to Done. An agent or contractor receives the task ID, relevant blueprint sections, input files, expected output, and acceptance criterion. They should not receive an instruction to improvise the entire game from a one-line prompt.

16. Verified platform references

Sources checked 27 October 2026. Version/availability claims are time-sensitive; recheck at the corresponding production gate. All numeric gameplay values, schedules, budget caps, and player-test thresholds elsewhere in this document are design choices or estimates.

- [S1] Godot Windows stable download: https://godotengine.org/download/windows/
- [S2] Godot high-level multiplayer: https://docs.godotengine.org/en/stable/tutorials/networking/high_level_multiplayer.html

- [S3] GodotSteam source: https://github.com/GodotSteam/GodotSteam ; multiplayer peer source: https://github.com/kargames/SteamMultiplayerPeer . A compatible release pairing is not established by this document.
- [S4] Steam Networking: https://partner.steamgames.com/doc/features/multiplayer/networking
- [S5] Steam Cloud: https://partner.steamgames.com/doc/features/cloud
- [S6] Steam review process: https://partner.steamgames.com/doc/store/Review_Process
- [S7] Steam Next Fest eligibility and workback: https://partner.steamgames.com/doc/marketing/upcoming_events/nextfest
- [S8] Steam store graphical assets: https://partner.steamgames.com/doc/store/assets/standard
- [S9] Steam Direct release waiting periods: https://partner.steamgames.com/steamdirect

The prior concept discussion used PEAK and PowerWash Simulator as adjacent audience examples. Their success is not treated as evidence that this game will sell, and no market size, revenue, or conversion estimate is included here.

2 / Contract design sheets

Version 1.0. These are authored production specifications. The included reference code checks C01 static power states only. The following timers, access sequences, remaining contract grids, and optional objectives still require Godot implementation and playtest validation.

Shared mission rules

Required final consumers must receive grid supply for 10 continuous seconds before departure. Required equipment is always provided. Temporary source ties must be disconnected before a second route to the same consumer or junction is energized. Access latches persist after a successful powered task. Optional deadlines affect bonus rewards, never mission solvability. The host advances all timers; pause freezes them only in offline solo mode.

Optional-service objectives are satisfied after the named load receives 10 continuous seconds of valid supply unless a sheet states a different duration. A completion latch is recorded for a service objective; final required supply is checked separately. A contract may not award the same optional objective twice. C01's sign is a training load excluded from optional-service objectives. In C01 the second star is granted on completion because there are no optional-service requirements.

All temporary cable decisions are socket and switch operations, not free-form wiring. A temporary generator has a designated placement pad near each eligible connection. Version 1.0 uses no fuel-depletion mechanic; the constraints are capacity, route access, equipment position, and service demand. Unused parts remain on the van; incorrect attempts cannot destroy the last required replacement.

District network baselines

District	Source / feeder limits	West required services	East required services	Required total
Alder Street	Grid 24; west/east feeder 12 each; generator 6	Apartments 4 + market 5 = 9	Pub 4 + apartments 4 + workshop 3 = 11	20 PU
Canal Yard	Grid 30; west/east feeder 14 each; generator 8	Laundrette 5 + diner 4 + row houses 4 = 13	Depot 4 + cold store 5 + pump kiosk 3 = 12	25 PU
Hill Road	Grid 28; west/east feeder 14 each; generator 6	Community hall 6 + bakery 4 + greenhouse 3 = 13	Flats 5 + telecom hut 4 + service garage 3 = 12	25 PU

PU are invented gameplay units. Internal service edges carry at least their load demand; feeder limits are the intended bottlenecks. Access pumps/control loads are separate consumers and do not remain connected at final extraction unless explicitly stated. Named alternative spurs are mutually exclusive with the ordinary supply route to their consumer. Do not add an alternative edge without the source/merge validation from the master blueprint.

C01 - First Shift

Introduce inspection, isolation, repair, generator placement, and access. Initial faults: JW fuse failed/tripped, east trunk broken. The 4 PU pump requires 12 continuous supplied seconds to expose the east repair. Known solution and exact graph are in the master blueprint and `data/first_shift.json`.

Required equipment: meter, repair tool, one fuse, one trunk replacement, cable reel, 6 PU generator. No optional service objective. Completion requires five required Alder consumers, pump access latch, equipment collection, and extraction. Sign test load 3 PU intentionally exceeds the east feeder if enabled; its first tutorial

trip does not count against the no-preventable-trip star.

C02 - Dinner Rush

Initial faults: west feeder connection broken and east trunk broken. Required grid demand remains 20 PU. Generator capacity 6 can temporarily serve market 5 or pub 4, but both would demand 9 PU. Add dedicated temporary sockets for those venues; the ordinary grid route must remain isolated while its temporary route is active.

Optional objectives: the market and pub must each accumulate 10 continuous supplied seconds before an authored 180-second opening-time deadline. The timer begins only after the briefing is dismissed and all players finish loading. Optional failure is reported immediately and never restarts the mission. Known solution: generator to pub; repair west connection and restore market from grid; both opening objectives latch; isolate pub temporary route, repair east trunk, then restore east from grid. There is no pump access gate in this contract. Test whether 180 seconds is readable and achievable before retaining it.

C03 - Roof Route

Initial faults: west fuse failed; normal east trunk broken and inaccessible behind a locked maintenance enclosure. A fixed ladder reaches a roof junction with a disabled alternate grid-to-JE feeder rated 12 PU. Restore JW to supply the ladder's access-light latch for 5 seconds, then use the ladder transition. Solo traversal requires no teammate.

Known solution: repair west fuse; latch access light; isolate JE; fit the supplied roof cable; enable alternate feeder; keep normal east trunk disconnected; restore JE. Final grid demand is 20 PU with east demand 11/12. The broken normal trunk does not have to be repaired if a valid grid bypass supplies every required consumer. Optional objective: power a separate 2 PU roof work light for 10 seconds using the spare generator; collect it before departure. Its optional completion is latched.

C04 - The Whole Block

Initial faults: west fuse failure and east trunk break; no pump gate. The pub sign is now a genuine optional 3 PU objective, and supplying it through JE would make that feeder 14/12 PU. Provide a separate disabled GRID-to-SIGN spur rated 4 PU, plus the normal JE-to-SIGN path. They cannot be active together.

Known solution: restore the two required branches to 20 PU; leave JE-to-SIGN off; install the direct sign spur while isolated; supply SIGN directly from GRID. Final source demand becomes 23/24 PU, while required east remains 11/12. Optional objective latches after 10 seconds. Keep the direct spur as installed infrastructure; do not require stowing a cable that is now the sign's permanent repair. This distinguishes permanent repair cable from portable deployed equipment.

C05 - Rising Water

Use Canal baseline. Both feeders have broken trunks behind independently flooded task areas. Two access pumps demand 5 PU and 4 PU; the 8 PU generator cannot run both simultaneously. Each pump requires 12 continuous supplied seconds and latches a separate drain condition. Neither access pump is a final service consumer.

Known solution: generator to west access pump; latch west drain; repair west trunk and restore 13 PU; move generator to east access pump; latch east drain; repair east trunk and restore 12 PU. Final grid demand 25/30. The inverse order must also work. Optional objective: restore the cold store within 180 seconds of first equipment pickup. Feedback shows the deadline and route decisions; never hide the clock in a menu.

C06 - Cold Chain

Use Canal baseline with west initially healthy and east trunk broken. A 4 PU auxiliary access pump must run for 12 seconds to expose the east repair. The cold store is 5 PU. The 8 PU generator cannot power cold store and pump together. Provide exclusive cold-store temporary and grid routes.

Optional objective: total cold-store time without any valid supply must remain below 120 seconds; the counter pauses while supplied and never resets. Known solution: temporarily supply cold store while preparing the repair; move the generator to the access pump; latch access; repair and restore the east grid route; reconnect cold store from grid before the outage allowance expires. Final 25 PU fits the grid. An alternate valid solution may sacrifice the cold-chain bonus. The garage/van always provides enough free parts to finish.

C07 - Wrong Label

Use Canal baseline. West starts supplied. East breaker is tripped and the installed feeder protection module limits the edge to 10 PU, despite a stale panel label reading 14. The meter reports the actual 10 PU component limit. Required east services demand 12 PU, so repeated resets cannot solve it.

Known solution: inspect the branch and component; isolate JE; replace the incorrectly rated module with the provided 14 PU part; close and verify east 12/14. This is an abstract module upgrade, not a claim about real fuse ratings. Optional objective: identify the mismatch through a meter inspection before the first attempted reset. Always allow recovery after a wrong reset; do not delete the replacement module. Final grid demand 25/30.

C08 - Split Supply

Replace Canal's single source with source A, capacity 18, and source B, capacity 16. A supplies west through a 14 PU feeder; B normally supplies east through a 14 PU feeder. B begins offline. A disabled 14 PU A-to-JE tie exists. All ordinary consumers retain Canal demands: west 13, east 12.

To access B's repair cabinet, supply the depot on the east branch for 10 continuous seconds; this latches the cabinet. A cannot supply all 25 PU. Known solution: disable west laundrette 5 and row houses 4, leaving diner 4; keep B-to-JE open; close A-to-JE tie, producing 16/18 total; latch depot access; repair B; open A-to-JE tie before closing B-to-JE; reenable west services. Final A load13/18 and B load12/16. Optional objective: keep the diner supplied throughout. Test the exact source transition for a one-tick phantom dual-supply bug.

C09 - Wind Warning

Use Hill baseline. West fuse is failed and east trunk is broken. The east repair has a roof route and a ground service-alley route. At mission second 60, show a gust warning; from second 90 through 130 the roof route closes to entry. Players already on the roof are offered an immediate safe return transition. The ground route remains available, taking approximately 20 seconds longer to traverse.

Known solution: repair west, then use the ground route at any time or the roof route outside the closure window; isolate and repair east; restore all 25 PU. Wind does not randomly break repaired equipment or force a fall. Optional objective: restore a 2 PU work-light access load for 10 seconds at a sheltered pad. The spare generator can supply it. Route-timing values are targets subject to navigation testing.

C10 - Community Night

Use Hill baseline; east begins healthy and west fuse failed. Community hall's required demand is 6 PU, included in west total13. The optional event stage adds 3 PU; through west it would demand16/14. A direct GRID-to-STAGE spur rated4 exists but must be installed. GRID capacity28 can supply required25 plus optional3 exactly.

Known solution: restore west; inspect branch headroom; keep WEST-to-STAGE open; install/enable the direct stage spur; verify total source28/28 and west13/14. Optional objective: stage supplied for10 continuous seconds before the event's 240-second start. If missed, all required services can still finish. Lights, music, and a small static crowd silhouette provide the payoff without NPC scheduling.

C11 - Borrowed Power

Use Hill baseline with west supplied and east trunk broken. Telecom4 can be temporarily supplied by generator6. A separate garage cabinet control load4 requires10 supplied seconds to unlock the east repair; generator cannot supply both at once. Provide an exclusive WEST-to-TELECOM temporary bypass. West baseline13 plus telecom4 would exceed14.

Known solution: supply telecom with generator; shed greenhouse3 so west demand becomes10; open telecom's generator route and enable west bypass, giving14/14 west; move generator to garage control and latch access; repair east trunk while JE isolated; open telecom west bypass before closing east feeder; restore east12 and greenhouse3, leaving west13. Optional objective: no telecom interruption longer than5 seconds; transitions must be clearly signposted and compatible with solo interaction times. A missed objective never blocks repair. Final source25/28.

C12 - Lights Back On

Use Hill baseline. West protection module failed, east trunk broken, and east cabinet access requires a generator-powered4 PU drain/control load for12 seconds. Add a genuine optional finale beacon3 on a direct4 PU grid spur, mutually exclusive with a west feeder route that would overload west to16/14.

Known solution: restore west13; generator to access load and latch; repair east and restore12; keep west-to-beacon off and install direct beacon spur; total28/28 exactly; collect portable equipment; verify required services; extract. Optional objective: power finale beacon for10 seconds. On successful verification, play a2-second staggered district lighting sequence and a short radio subtitle. Keep input active. The result screen completes the campaign and offers replay/remaining optional objectives rather than claiming an endless campaign.

Contract authoring checklist

For each contract instantiate stable nodes/edges, required consumers, optional-objective records, exclusive ties, route predicates, timer triggers, allowed recovery actions, and provided tools/parts. Reproduce the known solution with both solo and group input. Also test a plausible wrong route, missed optional deadline, item recovery, disconnect, and save/reload during each latch or supply transition. Store the observed solution trace and exact content revision with the acceptance report. The design sheets alone do not qualify a shipped contract.

3 / UI screen specifications

Version 1.0. All pixel measurements use a 1920x1080 reference canvas. Containers must reflow at other aspect ratios and UI scales. The five PNGs are implementation references, not screenshots of a built game.

Shared component contracts

- Primary button: amber fill, dark text, 56 px height, 16 px horizontal padding; one primary action per panel. Disabled state includes its reason in adjacent text or a focusable tooltip.
- Secondary button: panel fill with 2 px outline, 56 px height. Destructive actions use explicit verbs and a confirmation dialog naming the consequence.
- Status chip: icon + label + optional number. Unknown uses `?`, supplied uses check, temporary uses plug, fault uses broken-line symbol. Status never depends on color alone.
- Focus: 3 px high-contrast border, 4 px offset. Opening an overlay moves focus into it; closing restores the previous control. Modal dialogs trap focus until resolved.
- Toast: maximum two; 4 seconds for routine changes; persistent actionable errors stay until acknowledged or resolved. No toast covers the reticle or interaction prompt.
- Dialog: title, concrete cause, useful next action, Back. Clicking outside never dismisses a destructive or unsaved-change confirmation.
- Loading view: operation name, stage, elapsed time after 5 seconds, Cancel when safe. After 20 seconds show troubleshooting text and Back; do not promise an exact percent when work is unmeasurable.
- Scroll panels: visible position indicator, controller scrolling, no hidden primary actions below the fold. Every menu has a stable Back action.

S01 Boot and first-run setup

Sequence: minimal boot mark -> local settings load -> language selection on first run -> accessibility quick choices -> menu. Offer subtitle size, camera motion, flash reduction, input device check, and brightness preview. Skip applies defaults and all settings remain available.

Steam unavailable: show `Steam connection unavailable` with Retry and `Play solo offline`; online controls remain disabled with a reason. Do not repeatedly relaunch Steam. Unsupported save schema: offer `Open another profile` and diagnostics export; never silently erase a save. Startup corruption recovery tries one valid backup and reports that recovery occurred.

S02 Main menu

Left column: Continue, New campaign, Play with friends, Settings, Credits, Quit. Right card: selected campaign, latest contract, last played, version. Background: the depot/van scene at bounded quality, with an optional static background setting.

Continue is hidden when no valid campaign exists. New campaign opens a name field plus three local slots; overwriting requires confirmation with campaign name and last-played date. Focus defaults to Continue or New campaign. Online state is visible in the footer.

S03 Lobby

Top: session name, build compatibility, privacy. Left: four worker cards with local/host markers, readiness, connection state, and mute action. Right: contract summary, district image, required tools, estimated session length label. Bottom: Invite friend, Back, Ready/Unready; host sees Start shift.

Start requires all connected peers ready, compatible content, and finished loading. A late join uses `Joining` then `Ready`; it is never shown as an unexplained empty slot. Host can remove a peer through that peer's card with a confirmation. A mission already underway offers Join in progress if compatible. Error states cover lobby full, private/unavailable, protocol mismatch, content mismatch, Steam offline, invite expired, and host left.

Each offers Back and an appropriate Retry or Update instruction.

S04 Depot and contract board

Left: three district tabs and four contract cards. Center: selected contract briefing, required/optional objectives, introduced mechanic, previous result. Right: credits and loadout summary. Bottom: Back, Loadout, Start solo or Open lobby, according to current session.

Locked cards show the exact prerequisite. Completed cards offer Replay; replay does not overwrite an unrelated active campaign without confirmation. Guest profile completion can unlock earlier contracts but does not transfer another host's world state. Recommended next contract receives one marker; do not fill every card with badges.

S05 Loadout and upgrades

Two tabs: `Shift equipment` and `Depot upgrades`. Equipment slots show the item, assigned storage anchor, quantity, and whether supplied by the contract. A required-tool checklist is always visible. Missing optional gear never blocks Start. Purchases show current credits, cost, result, and remaining credits before confirmation.

The host validates equipment placement and contract requirements, but each player's credits and personal cosmetics are separate. Double-clicking Buy cannot charge twice. Sold-out/unlocked states replace the purchase action. Tool descriptions explain gameplay effects with units and limits. No placeholder speed bonus is sold.

S06 Field HUD

At x=80,y=70: district/contract line and up to three objectives in a 420 px-wide card. At x=1450,y=70: crew list. Center x=960,y=540: small reticle. At x=680,y=680: 560 px-wide interaction prompt with verb, binding, object, and missing condition. Bottom-left: four tool slots and carried equipment status. Bottom-right: one recent event plus Map/Ping hints.

First minute introduces one binding at a time after the player has used the previous action. The HUD tracks only information the player has measured or directly observed; the map can show unknown states. Looking at a teammate shows their name and current task, with privacy/mute settings honored.

S07 Network tablet

Full-screen panel with dark backdrop. Top: district, tab buttons `Network` and `Objectives`, close control. Left: interactive directed network map with stable IDs and icons. Right: selected branch evidence: source, measured demand/capacity, breaker state, fault notes, current worker, and Ping branch. Bottom: legend, zoom, center on self, and Back.

The map highlights energized paths, not hypothetical unmeasured routes as certain facts. Capacity displays `Measured: 11 / 12 PU`, or `Unknown - inspect junction`. During overload, pulse the path slowly or use a steady icon in reduced-flash mode. Clicking a node inspects information only; it does not toggle a remote physical switch. Controller left stick moves focus between nodes according to adjacency and right stick pans.

S08 Meter and repair view

Meter: small world tool with a mirrored UI readout for legibility. Readouts are supply present/absent/unknown, continuity healthy/broken/unknown, demand, limit, and breaker reason. An explanatory line connects evidence to the next action without giving away every later fault.

Repair overlay occupies the right third of the screen; player camera movement is constrained only while the panel interaction is active. Rows show diagnose, isolate, open, replace, close, test. Completed rows remain checked. Wrong conditions show a precise reason. Cancel preserves durable progress and releases the lock. Another worker's lock shows their name and allows Ping, not forceful takeover.

S09 Pause, options, and recovery

Solo pause suspends the simulation. Online menu explicitly shows `Shift continues online`. Actions: Resume, Settings, Recover player, Recover equipment, Leave shift. Recovery explains where the player/item will move. Host Leave warns that the session ends for everyone and shows checkpoint age. Guest Leave states that personal completed rewards are saved; it does not imply unfinished contract rewards exist.

Settings categories: Display, Graphics, Controls, Audio, Accessibility, Language. Graphics Apply keeps a revert timer for display changes. Device settings are local. Back returns to the exact prior tab and selection. Binding conflicts are visible and resolvable without a mouse.

S10 Results

Header: mission result and neighborhood restoration image. Center: required/optional services, elapsed time, trips, recoveries, equipment collected, and three star conditions. Right: reward receipt breakdown and new credit balance. Bottom: View network summary, Replay variant, Return to depot. Host chooses next world transition; guests can mark Ready or leave.

Rewards use a receipt status: Pending -> Saved. Reopening the results screen never reissues credits. A disconnected client receives its receipt when it reconnects to the same host session; host stores undelivered receipts. A client who never receives a valid completion receipt receives no invented offline reward. The base game does not use a central reward service.

S11 Connection and version errors

Always show the concrete operation and cause, such as `Could not join - host uses protocol 3, your build uses protocol 2`. Public strings omit raw peer IDs and network addresses. Actions include Retry, Return to lobby, Back to menu, or Copy diagnostics. Copy diagnostics strips account IDs and private paths by default.

On host disconnect, freeze interactions, show last durable checkpoint information and Return to menu. Reconnect attempts have bounded retries, visible progress, and Cancel. Do not spin indefinitely or show a successful lobby while its world has been destroyed.

S12 Credits, support, and diagnostics

Credits list contributors, fonts, assets, libraries, and required notices. Support view shows version, build ID, protocol, save schema, and a diagnostics export action. Exports include approved logs and manifest only; saves/chat/account identifiers require explicit selection. External support links ask before opening the browser. No telemetry upload occurs merely by viewing support.

UI implementation acceptance

Every screen must be usable at 1280x720, 1920x1080, and 1920x1200; UI scales 80%, 100%, 150%; English, pseudo-locale with 35% expansion, and Simplified Chinese; keyboard-only and controller-only. Verify first focus, Back, disabled reasons, error recovery, and retained selection for each. Store screenshots per matrix combination that changes layout. No final layout is accepted based only on the 1920x1080 mockups.

4 / Ordered implementation backlog

Each item is NOT_STARTED. Hours are initial focused-effort estimates. Actual evidence belongs on each completed item.

F01 - Create reproducible project

Role: engineering | Hours: 6 | Planned weeks: 1-2 | Depends on: none

1. Create repository and folders.
2. install Godot/export templates.
3. add input actions.
4. export blank Windows build.

Acceptance: A clean checkout imports and exports; manifest records engine and template hashes.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

F02 - Qualify Steam transport tuple

Role: engineering | Hours: 18 | Planned weeks: 1-2 | Depends on: F01

1. Install candidate GodotSteam and peer.
2. implement private lobby create/join.
3. record extension source revisions and binary hashes.

Acceptance: Two licensed accounts on separate real networks join; tuple locked; compatibility is measured.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

F03 - Prove host command authority

Role: engineering | Hours: 12 | Planned weeks: 1-2 | Depends on: F02

1. Spawn two players.
2. add one shared switch.
3. validate sender and sequence.
4. acknowledge durable revision.

Acceptance: Duplicate command changes switch once; invalid peer and entity IDs are rejected.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

F04 - Handle connection failures

Role: engineering | Hours: 8 | Planned weeks: 1-2 | Depends on: F03

1. Add Steam-offline solo mode.
2. version handshake.
3. disconnect return.
4. bounded reconnect.

Acceptance: Wrong version fails with a reason; reconnect and host-leave paths never hang.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

F05 - Establish visual tokens and block kit

Role: art_ui | Hours: 8 | Planned weeks: 1-2 | Depends on: F01

1. Create palette/font styles.

2. 4 m architectural modules.
3. interaction dimensions.
4. simple worker silhouette.

Acceptance: Kit imports at 1 unit per meter; HUD tokens match screen specification.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

F06 - Foundation gate and export evidence

Role: engineering | Hours: 6 | Planned weeks: 1-2 | Depends on: F04

1. Export internal build.
2. run real-network switch test.
3. log machine/network setup and exact hashes.

Acceptance: Real two-PC test passes; all unresolved failures recorded before grid integration.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G01 - Implement pure grid evaluation

Role: engineering | Hours: 14 | Planned weeks: 3-5 | Depends on: F01

1. Port reference graph rules to typed GDScript.
2. separate active topology from presentation.
3. stable ID order.

Acceptance: Reference scenarios produce equivalent canonical state; cycles and dual supplies rejected.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G02 - Add overload and reset rules

Role: engineering | Hours: 10 | Planned weeks: 3-5 | Depends on: G01

1. Evaluate at 10 Hz.
2. accumulate 1.5 s overload.
3. trip protected branch.
4. validate reset before change.

Acceptance: Equality at capacity works; short transient does not trip; sustained overload trips once.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G03 - Create meter and evidence model

Role: engineering | Hours: 12 | Planned weeks: 3-5 | Depends on: G01

1. Track unknown versus measured readings.
2. add supply/continuity/load readouts and inspection command.

Acceptance: Uninspected panel never displays false zero; selected branch evidence explains a fault.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G04 - Implement interaction and repair states

Role: engineering | Hours: 14 | Planned weeks: 3-5 | Depends on: G03

1. Add ray selection.
2. locks.
3. isolation preconditions.

4. discrete repair transitions.
5. durable cancellation.

Acceptance: Live repair refused; disconnect releases lock; one replacement consumed exactly once.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G05 - Build generator and socket routing

Role: engineering | Hours: 10 | Planned weeks: 3-5 | Depends on: G02

1. Create anchored generator placement.
2. route toggles.
3. 6 PU source.
4. curved cosmetic cables.

Acceptance: Pump or pub works; both overload; unsafe source tie refused before mutation.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G06 - Implement versioned save skeleton

Role: engineering | Hours: 10 | Planned weeks: 3-5 | Depends on: G04

1. Save causes and IDs.
2. temporary write and backup.
3. schema checks.
4. recompute derived state.

Acceptance: Kill during write retains valid current or backup save; unknown schema is not erased.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G07 - Integrate commands and grid

Role: engineering | Hours: 8 | Planned weeks: 3-5 | Depends on: G02, G04, G05, F06

1. Route all local/remote actions through host.
2. emit revisioned state.
3. create canonical hash.

Acceptance: Two peers agree after repair, trip, reset, and reconnect.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

G08 - Grid toy gate

Role: engineering | Hours: 6 | Planned weeks: 3-5 | Depends on: G06, G07

1. Run reference cases, interruption cases, and remote two-player toy.
2. archive logs and build.

Acceptance: No blocker; timing and topology assertions pass; five-minute raw capture retained.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V01 - Graybox Alder Street

Role: art_ui | Hours: 24 | Planned weeks: 6-9 | Depends on: F05, G08

1. Block six services and rear loop.
2. label JW/JE.
3. place van and generator pads.

4. simplify collision.

Acceptance: All routes traversable solo/controller; first restoration visible from the van.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V02 - Author C01 objectives

Role: engineering | Hours: 16 | Planned weeks: 6-9 | Depends on: V01

1. Implement pump latch and repair access.
2. final 10 s verification.
3. extraction readiness and cancel.

Acceptance: Known solution completes; optional pub sign never blocks campaign progress.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V03 - Implement recovery routes

Role: engineering | Hours: 12 | Planned weeks: 6-9 | Depends on: V02

1. Return stranded player/item to van pads.
2. preserve repairs.
3. return disconnected carried equipment.

Acceptance: Out-of-bounds, lost item, and guest disconnect cannot make C01 unsolvable.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V04 - Implement four-player snapshot flow

Role: engineering | Hours: 20 | Planned weeks: 6-9 | Depends on: G07, V02

1. Add content handshake, chunked snapshot, revision barrier, queued deltas, reconnect admission.

Acceptance: Join during pump/repair converges to canonical grid state; no duplicate object spawns.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V05 - Implement movement sync

Role: engineering | Hours: 18 | Planned weeks: 6-9 | Depends on: V04

1. Add host validation, prediction/reconciliation, remote interpolation, anchored equipment smoothing.

Acceptance: Movement remains usable at 150 ms RTT; no client-authoritative placement or repair.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V06 - Implement pings and text communication

Role: engineering | Hours: 12 | Planned weeks: 6-9 | Depends on: V02

1. Add four context pings, radial phrases, short escaped chat, mute and rate limits.

Acceptance: Two testers complete C01 with voice disabled using pings alone.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V07 - Produce field and tablet UI

Role: art_ui | Hours: 18 | Planned weeks: 6-9 | Depends on: V01

1. Build containers, context prompt, tool bar, crew list, graph navigation, icons and focus.

Acceptance: Map inspects and pings only; controller focus complete; 150% scale usable.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V08 - Implement personal reward receipts

Role: engineering | Hours: 10 | Planned weeks: 6-9 | Depends on: V02, G06

1. Assign contract instance IDs.
2. persist deduplication.
3. host receipts queue.
4. guest completion unlocks.

Acceptance: Replay/reconnect results do not duplicate credits; separate campaigns remain separate.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

V09 - Playable slice gate

Role: engineering | Hours: 8 | Planned weeks: 6-9 | Depends on: V03, V05, V06, V07, V08

1. Complete ten solo/group C01 runs including four players.
2. inspect logs and final state.

Acceptance: No mission blocker; all recoveries work; exact exported build retained.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A01 - Create art benchmark street

Role: art_ui | Hours: 28 | Planned weeks: 10-12 | Depends on: V09

1. Replace one building cluster with final kit.
2. author dark/temp/restored states.
3. tune navigation fill.

Acceptance: Three states readable; visual target runs on benchmark hardware before kit expansion.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A02 - Implement restoration audio and effects

Role: art_ui | Hours: 16 | Planned weeks: 10-12 | Depends on: A01

1. Add generator/pump/tool sounds.
2. bus volumes.
3. window recovery sequence.
4. reduced-flash variant.

Acceptance: Restoration readable without audio; volume balanced; flashes reducible.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A03 - Add local measurement events

Role: engineering | Hours: 10 | Planned weeks: 10-12 | Depends on: V09

1. Log objective timestamps, idle intervals, first repair, trips, hint use, disconnect, completion.

Acceptance: No raw account IDs in analytics export; event schema version recorded.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A04 - Create onboarding and core accessibility

Role: art_ui | Hours: 16 | Planned weeks: 10-12 | Depends on: V07

1. Introduce controls incrementally.

2. add UI scale, camera motion and flash controls.
3. hint ladder.

Acceptance: First repair discoverable; all essential status includes text/icon.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A05 - Run external slice study

Role: qa_product | Hours: 18 | Planned weeks: 10-12 | Depends on: A02, A03, A04

1. Recruit six outside pairs and three solo sessions.
2. observe without coaching.
3. capture consented evidence.

Acceptance: Report shows denominators, recruitment bias, timings, replay choices, and specific failure causes.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A06 - Revise failed mechanics and throughput

Role: engineering | Hours: 14 | Planned weeks: 10-12 | Depends on: A05

1. Change meter/routes/tradeoffs based on observations.
2. rerun affected scenario and record engineering/art hours.

Acceptance: Comprehension and replay gate passed or expansion explicitly blocked.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

A07 - Product continuation gate

Role: qa_product | Hours: 6 | Planned weeks: 10-12 | Depends on: A06

1. Review evidence.
2. revise capacity forecast.
3. choose expand, revise, or stop.
4. freeze demo feature list.

Acceptance: Decision includes observed evidence, open risks, and revised estimate; no sales claim.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D01 - Finish demo art and tool readability

Role: art_ui | Hours: 24 | Planned weeks: 13-16 | Depends on: A07

1. Complete Alder facades/task interiors.
2. tool hands.
3. panel labels.
4. restoration lighting.
5. remove placeholders.

Acceptance: C01 screenshots representative; task objects recognizable at normal play distance.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D02 - Implement complete menu/depot/lobby loop

Role: engineering | Hours: 18 | Planned weeks: 13-16 | Depends on: A07

1. Build boot, campaign slots, lobby ready checks, loadout, pause, errors, results and diagnostics.

Acceptance: All twelve screen groups have usable normal/empty/error states and Back paths.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D03 - Apply localization pipeline

Role: art_ui | Hours: 18 | Planned weeks: 13-16 | Depends on: D02

1. Move strings into keyed resources.
2. add pseudo-locale.
3. commission/review Simplified Chinese UI terms.

Acceptance: No essential text baked into textures; long strings and Chinese glyphs fit.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D04 - Complete controller path

Role: engineering | Hours: 12 | Planned weeks: 13-16 | Depends on: D02

1. Bind actions and remap.
2. implement radial selection, graph focus, glyph switching, binding conflicts.

Acceptance: Demo completes from launch to results with no mouse or keyboard.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D05 - Export and profile demo

Role: engineering | Hours: 10 | Planned weeks: 13-16 | Depends on: D01, D02

1. Run benchmark route.
2. measure frame time, RAM/VRAM, repeated reloads.
3. tune rain/shadows/emission.

Acceptance: Measured device report attached; presets produce a readable stable scene.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D06 - Create Steam store creative

Role: art_ui | Hours: 22 | Planned weeks: 13-16 | Depends on: D01

1. Name screening.
2. capsule/library art to current templates.
3. six screenshots.
4. 45-60 s gameplay trailer.

Acceptance: Art matches game; no advertised unimplemented features; template dimensions verified.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D07 - Demo acceptance study

Role: qa_product | Hours: 12 | Planned weeks: 13-16 | Depends on: D04, D05

1. Run 20 fresh sessions, clean-machine installs, real invites, reconnect, controller and accessibility checks.

Acceptance: No P0/P1; all demo acceptance cases recorded; player feedback summarized.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

D08 - Stage demo and store readiness

Role: qa_product | Hours: 8 | Planned weeks: 13-16 | Depends on: D03, D06, D07

1. Configure actual app/demo branches.
2. review page and demo checklists.
3. select event only after stable demo.

Acceptance: Publishable package prepared; external publication separately recorded; no invented event deadline.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT01 - Build Canal Yard kit variant

Role: art_ui | Hours: 24 | Planned weeks: 17-24 | Depends on: D08

1. Reuse six archetypes.
2. add canal silhouettes, pump access rooms and cold store.
3. graybox before polish.

Acceptance: Routes and services distinguishable; collisions and task positions validated.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT02 - Build Hill Road kit variant

Role: art_ui | Hours: 24 | Planned weeks: 17-24 | Depends on: D08

1. Create hill loop, community venue, roof landings, wind gates.
2. reuse approved modules.

Acceptance: Controller navigation and alternate routes complete before detail pass.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT03 - Implement service priority contracts

Role: engineering | Hours: 22 | Planned weeks: 17-24 | Depends on: D08

1. Author Alder C02-C04.
2. add optional venue load, fixed ladder access and two-fault layout.

Acceptance: Each contract has known solution; no mandatory extra grind or two-person task.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT04 - Implement pump and cold-store contracts

Role: engineering | Hours: 24 | Planned weeks: 17-24 | Depends on: CT01, CT03

1. Author C05-C06.
2. add independent access latches and optional cold-chain grace timer.

Acceptance: No pump deadlock; lost optional bonus never blocks completion.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT05 - Implement evidence and supply contracts

Role: engineering | Hours: 22 | Planned weeks: 17-24 | Depends on: CT04

1. Author C07-C08.
2. misleading label and exclusive supply tie.
3. require meter evidence.

Acceptance: Safe topology rules hold; solutions depend on measured state rather than memorized label.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT06 - Implement final district contracts

Role: engineering | Hours: 24 | Planned weeks: 17-24 | Depends on: CT02, CT05

1. Author C09-C12.
2. telegraphed wind event, event loads, generator relocation and finale.

Acceptance: All four complete solo/online; no invisible random fault changes.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT07 - Implement bounded progression

Role: engineering | Hours: 16 | Planned weeks: 17-24 | Depends on: CT03

1. Add unlock board, personal credit purchases, five starter upgrades, completion star rules.

Acceptance: Essential tools always supplied; simultaneous purchase retries cannot charge twice.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT08 - Finish campaign assets and animations

Role: art_ui | Hours: 28 | Planned weeks: 17-24 | Depends on: CT01, CT02

1. Complete shared worker rig/hands.
2. required animations.
3. signs, tools, venue sound variants.

Acceptance: Every interactive asset has readable state and matching sound/accessibility fallback.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT09 - Implement curated replay templates

Role: engineering | Hours: 16 | Planned weeks: 17-24 | Depends on: CT06

1. Choose bounded seeded fault templates.
2. validate supply/access/tool availability.
3. retain solution traces.

Acceptance: Every shipped seed template solvable; a seed reproduces the same definition.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

CT10 - Content completion gate

Role: qa_product | Hours: 14 | Planned weeks: 17-24 | Depends on: CT06, CT07, CT08, CT09

1. Complete all twelve contracts and inspect asset licenses.
2. compare measured content throughput with forecast.

Acceptance: Twelve solution traces and defect records exist; no missing required tool or final overload.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

H01 - Harden saves and campaign migration

Role: engineering | Hours: 16 | Planned weeks: 25-28 | Depends on: CT10

1. Finish backup recovery, previous-version migration, interrupted checkpoint and account separation.

Acceptance: Corrupt/newer saves never silently overwritten; old supported saves preserve progress.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

H02 - Harden rewards and disconnect recovery

Role: engineering | Hours: 18 | Planned weeks: 25-28 | Depends on: CT10

1. Test results redelivery, undelivered receipts, guest leave, host quit and duplicate commands.

Acceptance: Exactly-once local receipt application; checkpoints and participant state stay coherent.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

H03 - Complete accessibility and language layouts

Role: art_ui | Hours: 20 | Planned weeks: 25-28 | Depends on: CT10

1. Run each screen at scale/aspect variants.
2. add subtitles, speaker labels, hold/toggle and focus fixes.

Acceptance: No critical clipping; all tasks usable without color/audio/voice dependence.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

H04 - Optimize exported stress route

Role: engineering | Hours: 20 | Planned weeks: 25-28 | Depends on: CT10

1. Profile four-player host.
2. enforce bounded lights/particles/audio.
3. reduce allocations.
4. verify reload cleanup.

Acceptance: Measured P95 and memory report meets target or explicit revised hardware target proposed.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

H05 - Integrate Steam Cloud and achievements

Role: engineering | Hours: 14 | Planned weeks: 25-28 | Depends on: H01

1. Sync profile/campaign/receipts only.
2. exclude local settings.
3. implement a small verified achievement set.

Acceptance: Two actual PCs sync progress; machine graphics settings stay independent.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

H06 - Feature-complete alpha gate

Role: qa_product | Hours: 14 | Planned weeks: 25-28 | Depends on: H01, H02, H03, H04, H05

1. Play full campaign solo and four-player.
2. freeze features.
3. triage all known defects.

Acceptance: No known save-loss issue; every release feature complete; all P0/P1 blockers assigned.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

B01 - Run network/device beta matrix

Role: qa_product | Hours: 18 | Planned weeks: 29-32 | Depends on: H06

1. Test 80/150 ms RTT, loss/jitter, four-player soak, 20 lobby transitions and separate networks.

Acceptance: Actual impairment configuration, account setup, build and state hashes recorded.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

B02 - Run input/localization beta matrix

Role: qa_product | Hours: 14 | Planned weeks: 29-32 | Depends on: H06

1. Test resolution/aspect/UI scale/languages/controller and remaps.
2. commission final terminology review.

Acceptance: No missing glyph or inaccessible essential action; full focus paths verified.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

B03 - Fix authoritative state and protocol defects

Role: engineering | Hours: 20 | Planned weeks: 29-32 | Depends on: B01

1. Close network race, lock, replay, snapshot and rate-limit defects.
2. add targeted regression cases.

Acceptance: All reproduced P0/P1 defects fixed and retested under the same conditions.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

B04 - Fix persistence and UI defects

Role: engineering | Hours: 16 | Planned weeks: 29-32 | Depends on: B02

1. Close save, reward, Cloud and navigation failures.
2. verify clean install and update path.

Acceptance: No P0/P1; selected P2 exceptions have recorded impact and owner.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

B05 - Polish feedback and launch art

Role: art_ui | Hours: 18 | Planned weeks: 29-32 | Depends on: B02

1. Balance audio, final signposting and UI hierarchy.
2. refresh representative store captures.

Acceptance: Only shipped gameplay appears; reduced-motion and flash variants remain readable.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

B06 - Release-candidate beta gate

Role: qa_product | Hours: 12 | Planned weeks: 29-32 | Depends on: B03, B04, B05

1. Run complete acceptance matrix on exact candidate.
2. measure hardware target and revise store requirements.

Acceptance: Zero known P0/P1; performance and feature claims supported by candidate evidence.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

R01 - Audit Steam configuration and claims

Role: qa_product | Hours: 8 | Planned weeks: 33-36 | Depends on: B06

1. Check depots, executable, ownership/demo packages, languages, tags, support and content survey.

Acceptance: Advertised features exist; actual release candidate downloads and launches.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

R02 - Finalize launch creative and support folder

Role: art_ui | Hours: 14 | Planned weeks: 33-36 | Depends on: B06

1. Finalize trailer/screens/capsules.
2. credits/license notices.
3. known-issues and player help text.

Acceptance: No placeholder art or missing notice; all asset sizes match current Steam templates.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

R03 - Submit store and build reviews

Role: qa_product | Hours: 8 | Planned weeks: 33-36 | Depends on: R01, R02

1. Complete checklists.
2. submit with at least seven business days buffer.
3. track Valve feedback.

Acceptance: Approvals documented; requested corrections verified; no assumed automatic approval.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

R04 - Prepare reproducible release build

Role: engineering | Hours: 10 | Planned weeks: 33-36 | Depends on: R03

1. Export from tagged revision.
2. generate hashes/manifest.
3. remove debug tools.
4. retain rollback candidate.

Acceptance: Clean export matches intended content; no private logs/secrets or debug bypasses.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

R05 - Perform final release smoke

Role: qa_product | Hours: 10 | Planned weeks: 33-36 | Depends on: R04

1. Install from Steam on clean PCs.
2. invite, complete, save, Cloud sync, offline solo, controller path.

Acceptance: Exact candidate passes critical matrix; final signoff names build hash and remaining P2 issues.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

R06 - Stage launch and support rotation

Role: qa_product | Hours: 6 | Planned weeks: 33-36 | Depends on: R05

1. Confirm waiting periods and approved page.
2. stage chosen date.
3. assign crash/save/network triage.
4. retain prior build.

Acceptance: Concrete release checklist complete; publication is a recorded operator action.

Evidence: revision/build identifier, acceptance result, and relevant logs/screenshots.

5 / Acceptance test matrix

All game/Steam/device tests are PLANNED_NOT_RUN. Only the separately reported Python reference tests have been executed for this package. A reference scenario cannot pass an engine or network test.

ID	Severity	Steps	Expected result	Method
SIM01	P1	Load C01 initial graph; evaluate	No supplied consumers; no overload	Reference + Godot unit
SIM02	P1	Repair/reset west junction	APT_W and MARKET supplied; grid demand 9 PU	Reference + Godot unit
SIM03	P1	Power pump only from generator	Generator demand 4/6 PU; access timer can run	Reference static + Godot timer
SIM04	P1	Connect pump and pub to generator	8/6 PU overload; timed TEMP trip at 1.5 continuous seconds	Reference static + Godot timer
SIM05	P1	Enable pub sign after all required grid repairs	East edge 14/12 PU overload; grid source demand 23/24 PU	Reference + Godot unit
SIM06	P1	Connect two energized PUB sources	Command refused before mutation; map explains conflicting tie	Reference + Godot integration
SIM07	P1	Create a cycle/source backfeed/invalid ID	Definition or command rejected with concrete diagnostic	Reference + Godot unit
SIM08	P1	Set demand equal to capacity	Load remains supplied; no overload timer	Reference + Godot unit
SIM09	P1	Overload 1.4 s then remove optional load	Timer resets; no delayed spurious trip	Godot deterministic timer
SIM10	P1	Try reset while still overloaded	Reset refused; no rapid trip loop	Godot unit
SIM11	P1	Reverse node/edge input order	Canonical supplied state and capacity output unchanged	Reference + Godot unit
SIM12	P1	Attempt broken east repair before pump threshold	Access refused; valid route opens only after 12 continuous supplied seconds	Godot integration
SIM13	P1	Stop generator after pump completion	Drain remains latched; mission remains solvable	Godot integration
SIM14	P1	Lose supply during final verification	10 s verification resets; no premature completion	Godot deterministic timer
NET01	P1	Two clients request same repair lock simultaneously	Exactly one lock owner; other client gets worker information	Two real clients
NET02	P1	Duplicate durable repair command	Replacement consumed once; one state transition	Network integration
NET03	P1	Client sends out-of-range or wrong-inventory request	Host rejects; no world state mutation	Network integration

ID	Severity	Steps	Expected result	Method
NET04	P1	Join during pump drain and east repair	Snapshot plus queued deltas converges; canonical hash agrees	Real remote join
NET05	P1	Disconnect while carrying generator/holding lock	Item returns to valid pad; lock clears; remaining crew can finish	Real remote disconnect
NET06	P1	Reconnect same participant before extraction	One actor; correct equipment state; no duplicate reward eligibility	Real remote reconnect
NET07	P1	Host leaves or loses connection	Guests receive clear end screen; return works; checkpoint retained	Real host shutdown
NET08	P1	Wrong protocol/content revision invite	Join rejected before spawning with update reason	Two versioned exports
NET09	P1	150 ms RTT, 3% loss, up to 30 ms jitter	Interaction state converges; movement usable; no lost durable changes	Controlled impairment
NET10	P1	Four players run 60 min; 20 lobby transitions separately	No accumulating actors, unreleased locks, or session hangs	Exported soak
SAVE01	P0	Kill process during checkpoint write	Current or previous valid save loads; progress loss bounded by checkpoint	Fault injection
SAVE02	P0	Truncate current save	Backup recovery visible; no silent reset	Fault injection
SAVE03	P0	Open unsupported newer save schema	Refuse incompatible load; preserve bytes; offer another profile	Save integration
SAVE04	P1	Replay/reopen same completion receipt	Credits and completion applied once	Unit + integration
SAVE05	P1	Guest completes C08 then hosts own session	Personal board unlocks through C08; unrelated campaign world not overwritten	Two profiles
SAVE06	P1	Sync actual account across two PCs	Progress arrives; graphics/bindings/logs not synced	Steam Cloud real PCs
UX01	P1	Launch and complete C01 controller-only	Every focus/action reachable; bindings and glyphs correct	Human controller run
UX02	P1	Check all twelve screens at 720p and 150% UI	No essential clipped text/action; Back and focus restore work	Layout matrix
UX03	P1	Use pseudo-locale and Simplified Chinese	35% expansion fits; glyphs present; terms consistent	Localization review
UX04	P1	Disable audio; ignore color; no voice chat	Essential instructions and status usable through text/icons/pings	Accessibility run
UX05	P1	Disable motion/flashes; remap controls	All tasks remain completable with chosen settings	Accessibility run
UX06	P2	Fail Steam init or invite into full/private lobby	Specific cause, bounded wait, Retry/Back/offline option as relevant	Failure-path run

ID	Severity	Steps	Expected result	Method
MIS01	P1	Run C01-C12 solo and four-player	Known solution completes; required tools provided; optional failure nonblocking	Campaign matrix
MIS02	P1	Try all approved seeded variants	Deterministic definition; no missing supply/access/equipment route	Content validator + sampled play
MIS03	P1	Recover player and all equipment from adverse locations	No item loss, irrecoverable route, or mandatory multiplayer carry	Recovery matrix
MIS04	P1	Exit extraction zone during countdown	Countdown cancels; no forced partial departure	Multiplayer integration
PERF01	P1	Four-player host exported stress route on target PC	Record P95 frame time and VRAM; meet target or revise claim before release	On-device profiler
PERF02	P1	Reload mission five times	No sustained resource growth after warm-up; bounded particle/audio counts	On-device profiler
REL01	P0	Clean install from actual Steam candidate branch	Launch works; no missing extension/runtime/content	Clean PC
REL02	P1	Exit Steam; start offline solo profile	Offline path works; online disabled with reason; no profile corruption	Offline export run
REL03	P1	Audit feature flags and published store claims	Only implemented and verified features advertised	Release checklist
REL04	P1	Compare final build manifest and test report	Exact tested revision/content/extension hashes match staged candidate	Manifest audit

Evidence record

For each execution record test ID, build and content revision, engine/extension tuple, hardware/driver, input device, network impairment settings where relevant, exact reproduction steps, observed result, pass/fail, and evidence path. Retest the same failure condition after a fix.

P0/P1 failures block release. Tests with reference coverage must still run against the Godot implementation. Tutorial hints, accessibility settings, and saved checkpoints must be tested as distinct state changes, not assumed to work because the normal path passed.

6 / Content and asset register

Production assets are not yet created. Record creator/source/license, source revision, approval, and import evidence for every item. Counts describe families or variants as named, not a promise that each entry is one file.

ID	Family	Count	Implementation
ENV01	4m modular structural kit	11	Blockout -> collision -> UV/material -> LOD/import -> on-device check
ENV02	Facade archetype palettes and signs	6	Distinct silhouettes; no essential text baked into shared textures
ENV03	Alder service locations	6	APT_W, MARKET, PUB, APT_E, WORKSHOP, PUMP_HOUSE
ENV04	Canal service locations	6	Laundrette, diner, depot, cold store, row houses, pump kiosk
ENV05	Hill service locations	6	Community hall, bakery, flats, greenhouse, telecom hut, service garage
PRP01	Junction panel variants	3	Closed/open/isolated/fault/tripped states; matching collision and repair anchors
PRP02	Portable generator and cart	2	Solo carry, placement anchors, running/overload state; no rope physics
PRP03	Meter, cable reel, fuse case, toolbox, work light, pump	6	Tool use anchors, icons, state materials and audio hooks
PRP04	Van exterior and equipment station	2	No drivable vehicle system; boarding/loading transition
PRP05	Barrier, drain, bin, debris and sign families	5	Simplified collision; no player snags
CHR01	Shared worker body and rig	1	Four helmet/vest palettes; neutral face; no facial animation
CHR02	Worker/tool animation clips	12	Idle, walk, sprint, crouch, carry, interact, tester, replace, switch, stumble, recover, ping
UI01	Twelve screen groups	12	Normal/empty/error/loading/disabled/focus states; localization keys
UI02	Tool/status/ping icons	24	SVG source; color-independent silhouettes; 24/32/48px exports
FX01	Rain, drip, gust, reduced-flash arc, water plane	5	Bounded local emitters and scalable quality; no fluid solver
AUD01	Core sound families	18	Rain, sheltered rain, wind, generator idle/load, tester, switch, fuse, trip, pump start/drain/stop, cable, three footsteps, ping, restoration
MKT01	Trailer, six screenshots, store/library art set	1	Current Steam templates; representative footage; final counts follow official checklist

Required asset pipeline

1. Record creator/source/license.
2. Build blockout and gameplay anchors.
3. Approve silhouette and scale.
4. Create final geometry/materials.
5. Simplify collision and add required states.
6. Export/import with stable naming.
7. Check lighting/UI legibility.
8. Profile in exported build.
9. Mark approved and commit source revision.

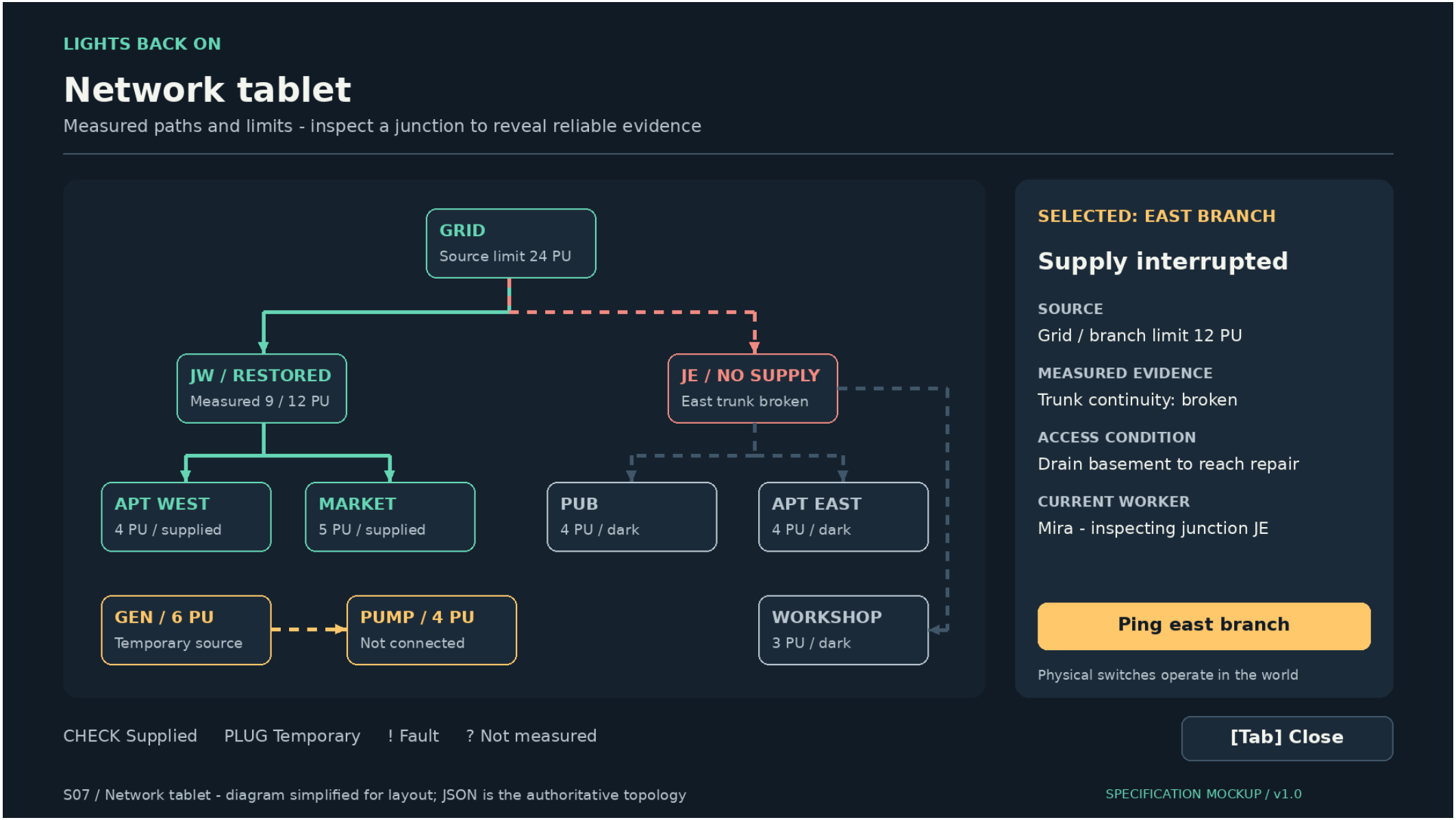
Version manifest

engine_version, engine_hash, export_templates_hash, steam_wrapper_revision, multiplayer_peer_revision, extension_binary_hashes, steamworks_runtime_version, protocol_version, save_schema_version, content_revision, git_commit, build_id

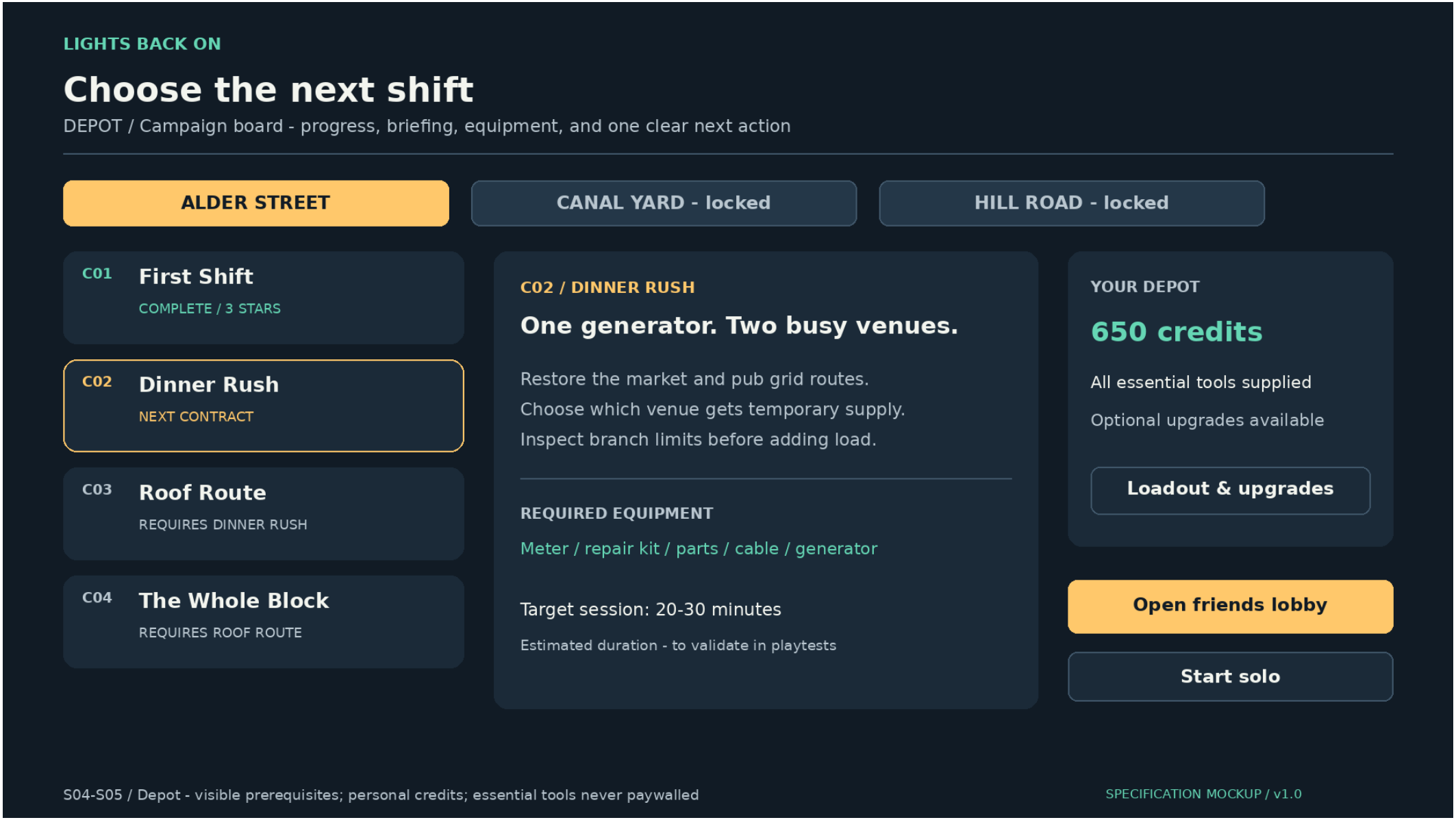
7 / Visual implementation references



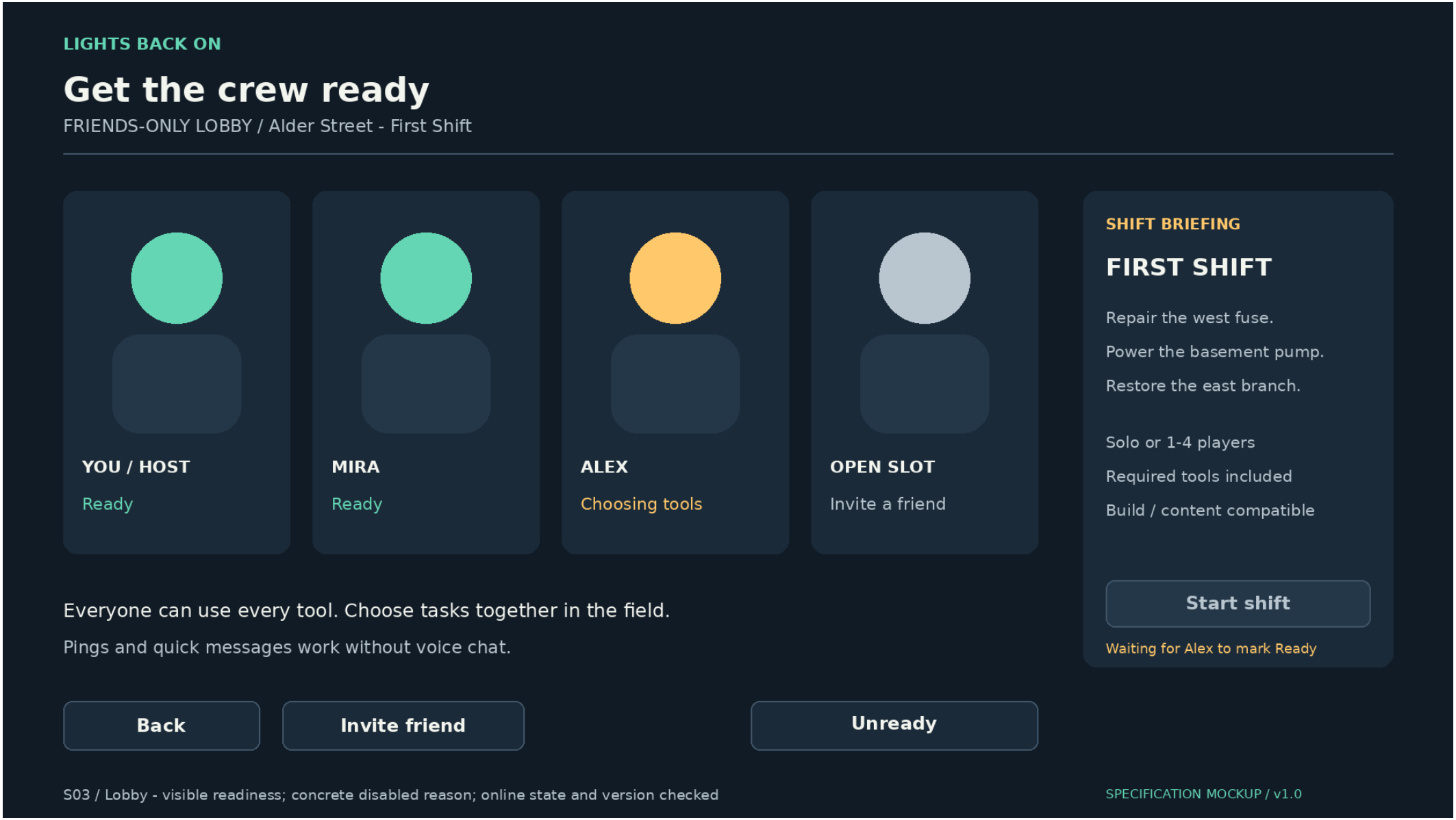
Layout specification; full-resolution PNG and behavioral specification are included in the editable package.



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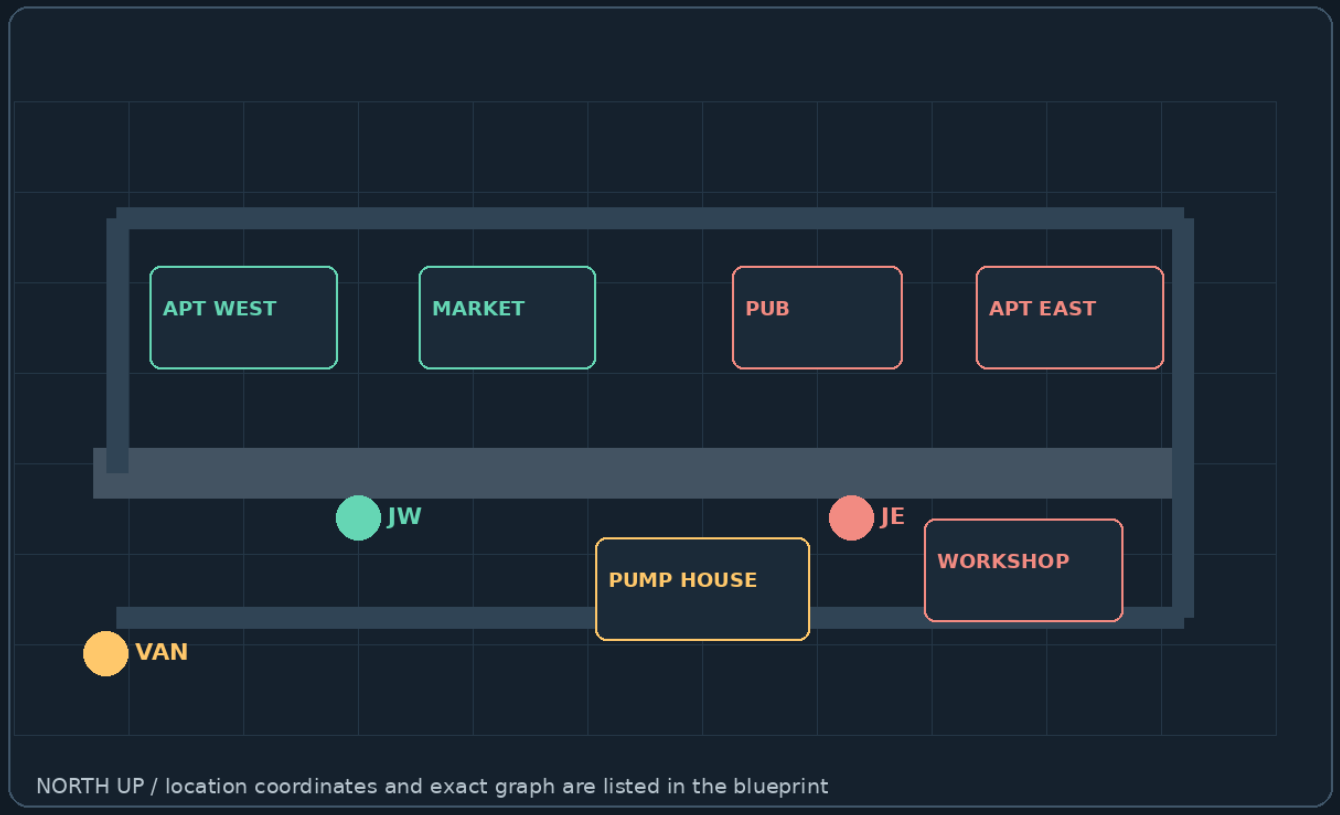


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LIGHTS BACK ON

Alder Street / first district

110 x 70 m proposed playable envelope - arrangement reference, not a surveyed plan



NORTH UP / location coordinates and exact graph are listed in the blueprint

ROUTE AND SIGHTLINE RULES

01 / ORIENTATION
Van faces the first lit facades.

02 / PARALLEL TASKS
West repair, pump setup, east test.

03 / ACCESS GATE
Drain basement to reach east trunk.

04 / CO-OP FLOW
Rear alley avoids one narrow route.

05 / RECOVERY
Valid storage pads near every task.

06 / READABLE SCALE
4 m modules; 2.4 m main routes.

LEVEL PLAN / One street loop; six services; compact interiors; no open-world driving

SPECIFICATION MOCKUP / v1.0

Layout specification; full-resolution PNG and behavioral specification are included in the editable package.